GazeSync: Reproducible Synchronization of Screen Interaction and Multi-Source Eye Tracking

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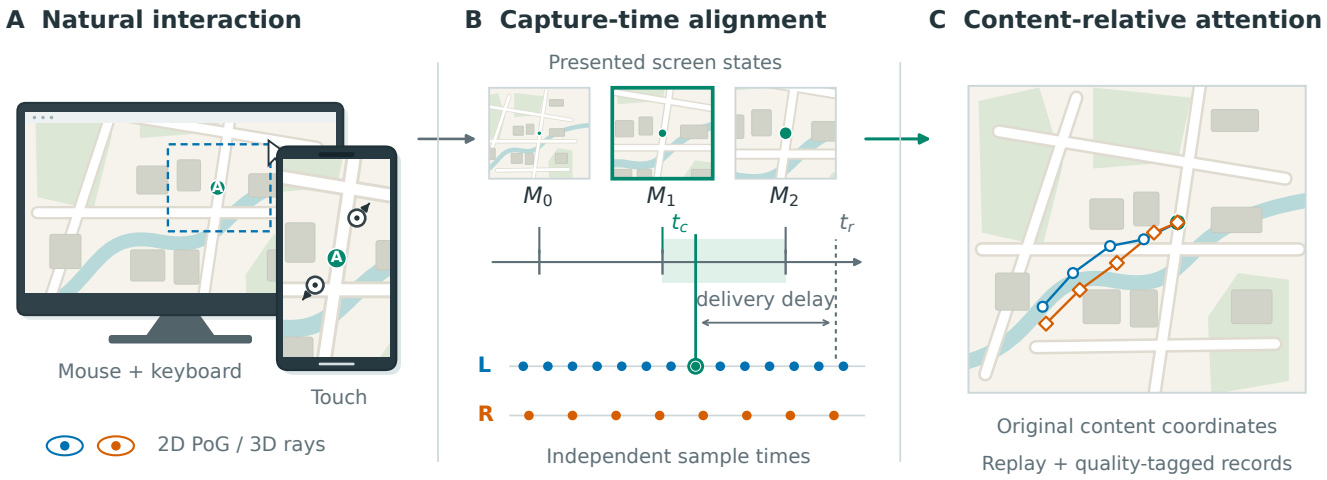


Figure 1: GazeSync links natural interaction to content-relative attention. (A) Desktop and mobile views change through mouse, keyboard and touch input. (B) Each eye retains independent timestamps; the screen state at capture, M_1, is selected despite later delivery. (C) Gaze is reconstructed on the original content, with raw records and quality decisions retained. The map views and sample paths are schematic.

Abstract

A gaze sample and a zoom event can arrive together even though they describe different screen states. Recovering attention on manipulated content therefore requires explicit temporal and spatial provenance. We extend GazeSync into a standalone Python and TypeScript framework for desktop and mobile interaction, 2D points of gaze, 3D eye-origin rays, and independent monocular or binocular streams. The framework separates capture, input, presentation and receipt time; joins each observation to a surface-specific presentation history; and preserves uncertainty, eye identity, geometry and missing data in a replayable record. SDK-independent adapters normalize model and tracker records without assuming equal acquisition rates or manufacturing confidence. A reproducible synthetic matrix spans 78 configurations and 107,070 observations, including 30–240 Hz displays, 30–2,000 Hz gaze and nine viewport sizes. Both implementations select the reference capture frame in every case; maximum cross-language viewport disagreement is 4.55×10^{-13} logical pixels. These are numerical consistency results, not physical tracking-accuracy estimates. We provide an inspectable synchronization contract, diagnostic guidance, simulation fixtures and isolated distribution tooling, and identify the device measurements needed before making end-to-end accuracy claims.

Keywords

eye tracking, temporal synchronization, pan and zoom, binocular gaze, reproducibility, research software

1 Introduction

A person zooms into a map landmark, rotates the map, drags it, and then resets the view. A screen-coordinate gaze point can refer to different content locations during that sequence. A delayed tracker callback adds another ambiguity: the view visible when the callback is handled may differ from the view at exposure. On a phone, simultaneous finger movement and changing display pose introduce further spatial dependencies. Researchers need to distinguish these effects before interpreting a trace as attention to a particular feature.

The original GazeSync introduced mobile gaze and image-transform logging with content-relative reconstruction [1]. Within the broader mobile-gaze pipeline of sensing, estimation and interaction reviewed by Lei et al. [2], this extension addresses the association of gaze observations with rendered content. It defines how a renderer and heterogeneous gaze sources exchange timing, geometry and validity, then provides matching Python and TypeScript implementations. The target is instrumentation of an application owned by the researcher: the library does not observe

Table 1: Implemented contracts and their verification. Device calibration and physical validation remain host responsibilities.

Layer	Contract and evidence
Interaction	Wheel, drag, box and keyboard; pinch, twist and free movement. *Evidence:* gesture and browser regressions.
Signals	2D PoG, 3D rays and direction-only records; independent or combined eyes. *Evidence:* analytic and source fixtures.
Alignment	Capture-time presentation history, uncertainty and quality flags. *Evidence:* conformance and exact replay.
Integration	SDK-independent model and tracker records. *Evidence:* synthetic protocol fixtures.
Distribution	Python and TypeScript, zero runtime dependencies. *Evidence:* clean package installations.

arbitrary operating-system windows or infer unrecorded visual states.

Our contribution has three parts. First, a capture-time contract preserves the screen state and source provenance needed to reconstruct content-relative gaze. Second, independent eye channels, source adapters and actionable diagnostics make heterogeneous measurements explicit. Third, deterministic simulations, cross-language conformance and isolated distribution artifacts make the implementation inspectable and repeatable. The evidence is a software audit and synthetic validation; no new participant or physical-device study is claimed. This draft documents implementation 0.5.0 and recording schema 2.1. The development repository is https://github.com/GazeLei/GazeSync.

2 Background and scope

2.1 Relationship to the original GazeSync

The prior system used a Flutter/iOS application with an on-device gaze SDK and local logging server. Its callback pairing and nearest-state merge used a 50 ms tolerance. A three-participant formative study illustrated both useful content-relative reconstruction and calibration or compound-transform limitations [1]. Those participants and outcomes belong to the prior paper; they are not additional validation of this implementation.

The extension replaces the platform-specific merging assumption with explicit presentation intervals, source clocks and quality decisions. It adds desktop interaction, independent eye identity, metric ray projection, reusable adapters and a common record format across languages. Table 1 states the boundary between capabilities and evidence. A source can be accepted at many frequencies without establishing that a particular device exposes accurate capture timestamps at those frequencies.

2.2 Interaction and synchronization systems

Gaze-supported pan and zoom has been investigated as an interaction technique, including combining gaze with other input modalities [6]. DynamicRead evaluated gaze-based scrolling for handheld reading in sitting and walking conditions, finding differences in the techniques' robustness under mobility [3]. Our instrumentation observes manual interaction while retaining gaze for analysis; it does not assume that the gaze target is the pointer or pinch midpoint. A person may inspect a landmark while it moves, look at a stable screen location while dragging, or shift attention ahead of the next action. These scenarios motivate varied attention policies in the simulation; the scripted trajectories and their frequencies remain illustrative.

Lab Streaming Layer provides timestamped multimodal streams and network clock synchronization [7]. Such infrastructure can supply a transport or clock estimate to a GazeSync host. Our core instead specifies which rendered content state a mapped timestamp refers to, how coordinates are reconstructed, and what makes the result ambiguous. It neither implements a network clock estimator nor claims to replace multimodal acquisition infrastructure. Device-specific data contracts also remain consequential: Tobii display coordinates and metric user coordinates differ from Pupil scene-camera coordinates [8, 9]. Explicit normalization is therefore part of the measurement contract, not a cosmetic import step.

3 Design requirements

Record what changed on screen. Mouse, keyboard and touch events express intent. A renderer can coalesce many inputs into one display update or animate a single command across several frames. We record input events for explanation and presentation snapshots for reconstruction. A snapshot includes surface, layout epoch, content identity, viewport dimensions, the rendered affine matrix, timing provenance and optional registered display geometry. Layout changes and suspension are observable events.

Respect independent measurements. A binocular tracker can lose one eye, sample eyes at different times, or expose a fused output with different semantics. The record retains `left`, `right`, `combined` or `unknown` identity. Separate observations retain their timestamps and validity. The software does not synthesize a second eye from a combined estimate or treat vendor validity as a probability.

Keep disagreement visible. Missing history, uncertain clocks, incompatible coordinates and invalid samples must remain inspectable. A seemingly smooth trace can conceal an acquisition error. Raw data and quality flags are retained alongside derived coordinates; blinks and gaps break a plotted trace. Rates and thresholds are configurable study parameters rather than universal recommendations.

Keep adaptation local and testable. The synchronization core depends on neither a hardware SDK nor a learned model. Adapters convert explicit records at the edge, while a single finite-JSON contract carries the result. This permits hardware or inference changes without changing the content-reconstruction rules. Cross-language fixtures check that Python analysis and TypeScript acquisition interpret the same record consistently.

4 Temporal reconstruction

4.1 Four event times and a clock epoch

We distinguish input time t_i, presentation time t_p, source capture time t_s and session receipt time t_r. A frozen mapping for clock epoch e converts source time to session capture time:

$$t_c = b_e + r_e(t_s - a_e), \tag{1}$$

where a_e and b_e are corresponding anchors and r_e is the clock-rate ratio. Canonical times are monotonic milliseconds. Browser timing follows the monotonic clock model of High Resolution Time [13]; an epoch wall clock cannot be substituted without a measured mapping.

The host supplies clock uncertainty and optional drift uncertainty. The latter grows with distance from the anchor; conversion also includes a conservative floating-point term. A revised fit or source restart requires a new epoch. These are supplied engineering bounds, not a fitted statistical confidence interval. The framework does not estimate clock parameters from network arrival or assume symmetric one-way transport delay. Receipt-only sources remain recordable but are excluded from capture-time attention analysis.

4.2 Presentation intervals and ordering

For the surface-specific history $\mathcal{H}_s$, alignment selects

$$j^* = \underset{j \in \mathcal{H}_s:\, t_j^p \le t_c}{\arg\max}\ (t_j^p, \text{sequence}_j). \tag{2}$$

The last submitted snapshot wins a timestamp tie. Each snapshot persists until its successor. Geometry is not interpolated through an unobserved animation, reset or content replacement. The host must record every relevant intermediate rendered state, or explicitly assert change-only presentation logging.

A call to `flush(h)` is a watermark: it declares that all presentations through h have already been submitted. Insertion at or before that watermark is rejected. Samples normally wait for the greater of the configured reorder window and maximum accepted clock uncertainty. Ready samples are sorted by capture time within each flush. Later arrivals remain identifiable even when retained history permits valid offline reconstruction. Impossible future times and excessive uncertainty are quarantined promptly, preventing indefinite pending-buffer growth.

The interval $[t_c - u_c, t_c + u_c]$ is checked against visual changes. If it crosses a relevant change, the output is marked ambiguous. Boundaries that repeat identical geometry do not create false ambiguity. Histories are indexed per surface; retention preserves the newest state of each active surface, so frequent updates on one display do not evict another display's only state. A required older state may still expire, producing an explicit history flag.

4.3 Why timing errors become spatial errors

For a viewport observation p_v and rendered matrix $M(t)$, content gaze is $M(t)^{-1}\widetilde{p}_v$. Away from discontinuities, a small timing displacement Δt yields a first-order spatial error proportional to

$$\left\| \frac{dM(t)^{-1}}{dt} \widetilde{p}_v \right\| |\Delta t|. \tag{3}$$

Fast zoom, rotation or translation can therefore amplify a modest delivery delay. At a reset or layout discontinuity this derivative approximation does not apply; the correct presentation interval must be selected explicitly. Figure 2 illustrates this mechanism with known synthetic timing. The receipt-time baseline is an intentionally incorrect association policy, not a benchmark against every strategy used by existing systems.

5 Spatial reconstruction and eye identity

5.1 Named frames and distinct origins

We use column vectors and write T_{BA} for a change of coordinates from frame A to frame B. Each 3D observation and plane carries a `frameId`; equality of identifiers is necessary, but does not establish calibration by itself. The host must document the origin, axis directions, handedness, units and time reference of each frame. Table 2 distinguishes origins that are easy to conflate during integration.

The camera's optical centre differs from its image principal point, which is an intrinsic calibration parameter. A gaze ray's origin is the measured eye centre expressed in the chosen frame, generally not that frame's origin. Zero is valid for an eye-local origin only when its relation to the display is known. Vendor conventions also differ: Tobii UCS is centred on the tracker front surface, with user-right/up/toward-user axes and millimetre units [8]. Pupil's scene-camera frame and individual eye-camera frames are distinct [12]. The adapters retain these identities and convert metric units explicitly.

5.2 Rigid transformations: points versus directions

The implementation accepts a row-major 3×3 proper rotation R_{BA} and translation t_{BA} in metres:

$$T_{BA} = \begin{bmatrix} R_{BA} & t_{BA} \\ 0^T & 1 \end{bmatrix}, \qquad p_B = R_{BA}p_A + t_{BA}. \tag{4}$$

Here t_{BA} is the source origin expressed in the target frame. The rotation must satisfy $R^TR = I$ and $\det R = +1$; reflections, scaling and shear are rejected. The helpers distinguish ray origins and directions:

$$O_B = R_{BA}O_A + t_{BA}, \qquad D_B = R_{BA}D_A. \tag{5}$$

Translation must not be applied to a direction, whose homogeneous coordinate is zero. Inversion gives $R_{AB} = R_{BA}^T$ and $t_{AB} = -R_{BA}^T t_{BA}$.

OpenCV extrinsics map world coordinates into camera coordinates, $p_C = R_{CW}p_W + t_{CW}$ [11]. Consequently the camera's world position is

$$C_W = -R_{CW}^T t_{CW}, \tag{6}$$

not t_{CW}. Eye origins must be transformed independently. A display-mounted camera may observe eyes at positive camera z while their gaze directions point back toward the display with negative z; this is geometrically valid. The `transformRay` and `transformPlane` helpers check named source frames, return new values, and never infer a missing eye origin. Python provides the same operations with snake-case names.

An eye-local ray with origin zero has $O_C^e = t_{CE_e}$ and $D_C^e = R_{CE_e}D_{E_e}$, independently for $e \in \{L, R\}$. Its measured registration supplies the nonzero camera-frame eye origin.

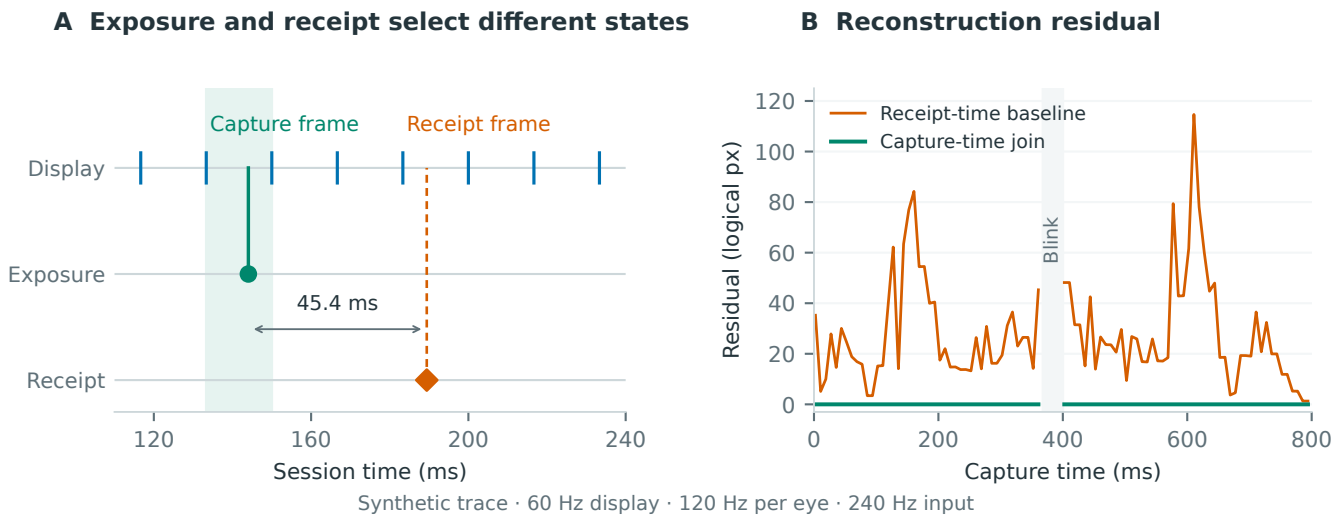


Figure 2: Observed timing in the representative synthetic trace. (A) Capture and receipt fall in different presentation intervals. (B) Receipt-time association introduces reconstruction residuals during manipulation; capture-time residuals remain at numerical scale. The blank interval denotes a blink. This is a software fixture, not a physical timing measurement.

Table 2: Coordinate origins and conventions. Eye labels are anatomical. The optical-camera axes shown here are an explicit example; each vendor frame requires its documented conversion.

Frame	Origin	Convention and units
Camera C	Optical centre of the calibrated camera	Example: $+x$ image-right, $+y$ down, $+z$ forward; metres.
Eye E_L, E_R	Each measured eye centre	Eye-local zero requires a measured transform into C or W; metres.
Reference W	A fixed, documented host reference	Here: active-display centre at the reference pose; metres.
Screen S	Active rectangle's top-left, excluding bezel	Right u, down v, normal $u \times v$; metres.
Image	Pixel origin specified by image calibration	The principal point (c_x, c_y) is distinct from the optical centre; image pixels.
Display D	Full active display's top-left	Axes follow the current display orientation; logical pixels.
Viewport V	Stimulus region's top-left, at (l, t) in D	Parallel display/viewport axes for the offset mapping; logical pixels.
Content I	Intrinsic origin of the untransformed content	The recorded matrix M maps I to V; content units.

A plane's top-left point transforms with R and t, while its right/-down axes transform with R only. A transformed plane remains marked static only if both the original plane and transformation explicitly declare time independence. Applications retain the original ray, calibration identifier, transform and pose time in sample metadata, so a frame conversion does not erase its provenance.

5.3 Screen plane and forward intersection

A registered active rectangle has top-left P, orthonormal right/down axes u, v and physical width/height w_m, h_m. Its local-to-reference frame is

$$T_{WS} = \begin{bmatrix} u & v & u \times v & P \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{7}$$

A marker-board pose may locate the marker centre rather than the active panel corner. A separately measured registration is then required: $T_{CS} = T_{CM} T_{MS}$. The host fits a rigid pose to the known rectangle and supplies its physical dimensions; noisy corner differences do not automatically form an orthonormal screen basis.

For a user-right/up/toward-user reference, $u = (1, 0, 0)$ and $v = (0, -1, 0)$, so the screen-local positive normal points behind the front face. Its sign does not impose a sign restriction on a valid gaze direction.

Normalize the nonzero direction to $d = D/\|D\|$. For $n = u \times v$, the crossing is

$$\lambda = \frac{(P - O) \cdot n}{d \cdot n}, \qquad Q = O + \lambda d. \tag{8}$$

A valid forward crossing requires $\lambda > 0$. Reversing n changes numerator and denominator together, preserving λ; a negative denominator alone is not a reason to reject the ray. Zero directions, $|d \cdot n| < 10^{-9}$, non-positive distance and non-finite results are rejected. Robust direction normalization avoids overflow for very large or small vector magnitudes. The reported distance is measured from the eye, rather than camera depth.

The normalized crossing within the plane rectangle is

$$\alpha = \frac{(Q-P)\cdot u}{w_m}, \qquad \beta = \frac{(Q-P)\cdot v}{h_m}. \tag{9}$$

The operation is a two-sided geometric crossing. For an opaque panel with a known front-facing right/down basis, $(O-P)\cdot(u\times v) < 0$ places the eye in front. The host must establish front-facing and occlusion visibility; an inactive surface is recorded explicitly. A geometric hit alone does not establish an unobstructed view of the content.

The infinite-plane crossing and the active rectangle are separate tests: $\alpha, \beta \in [0,1]$ identifies the finite rectangle. Off-display intersections are retained and flagged, including when an oversized application window would contain them. A successful aligned 3D hit records its metric frame ID as well as its position and distance. This operation does not infer scene fixation depth, vergence or an unmeasured camera-to-display transform.

5.4 Display, viewport and content coordinates

The geometric audit exposed a missing distinction between the physical rectangle represented by a plane and the logical viewport. If the plane describes the *full active display*, with logical dimensions W_D, H_D, then

$$p_D = (W_D\alpha, H_D\beta), \qquad p_V = p_D - (l,t), \tag{10}$$

where (l,t) is the viewport's top-left in display coordinates. This conversion is selected by `planeSpace: display` and requires explicit display layout. It matches the normalized-display PoG path.

If the plane instead describes the physical *viewport subrectangle*, then `planeSpace: viewport` gives $p_V = (W_V\alpha, H_V\beta)$ without subtracting the offset a second time. For an axis-aligned crop, the equivalent viewport plane can be constructed as

$$P_V = P_D + u\,w_m\frac{l}{W_D} + v\,h_m\frac{t}{H_D}, \tag{11}$$

$$w_m^V = w_m\frac{W_V}{W_D}, \qquad h_m^V = h_m\frac{H_V}{H_D}. \tag{12}$$

The two paths are equivalent when configured consistently. An ambiguous plane extent is rejected when the recorded viewport differs from the full display. These offset formulas assume parallel viewport/display axes. Outer application transforms must be resolved by the host; rotating the rendered content is handled separately by its affine matrix.

The content-to-viewport transform is

$$M = \begin{bmatrix} a & c & e \\ b & d & f \\ 0 & 0 & 1 \end{bmatrix}, \qquad \widetilde{p}_I = M^{-1}\widetilde{p}_V. \tag{13}$$

It includes intrinsic scaling, letterboxing, pivot translation and content rotation. Each accepted frame caches the inverse; scale-normalized conditioning checks reject singular transforms while permitting small, well-conditioned scales. Device pixel ratio is metadata, not an extra factor in Equation 10. Device-pixel measurements are normalized using their source resolution before conversion into logical units.

Canonical UI coordinates describe continuous rectangle edges from $(0,0)$ to (W,H), not integer array indices. Pixel-centre conventions must be converted once when required by a source. Bounds classification allows arithmetic residuals of 64 machine epsilons scaled by the coordinate extent, without changing the stored coordinates. This numerical allowance is not a tracking-accuracy margin.

5.5 A numerical origin and viewport check

The shared analytic fixture uses a 0.53×0.33 m panel centred at O_W, with $P_W = (-0.265, 0.165, 0)$ and camera centre $C_W = (0.04, 0.19, 0.015)$ m. It deliberately places the camera away from both panel and eye origins. With $R_{CW} = \mathrm{diag}(-1,-1,1)$, $t_{CW} = (0.04, 0.19, -0.015)$ m. The left eye at $O_W^L = (-0.032, 0, 0.6)$ becomes $O_C^L = (0.072, 0.19, 0.585)$. A centre-directed $D_C^L = (-0.032, 0, -0.6)$ transforms to $D_W^L = (0.032, 0, -0.6)$ and intersects at $Q_W = (0,0,0)$. The right eye is checked independently at $O_W^R = (0.032, 0, 0.6)$.

For a 1920×1080 logical display and 1000×600 viewport at $(100, 80)$, both eyes yield $p_V = (860, 460)$. The previous full-plane-to-viewport scaling incorrectly produced $(500, 300)$, a regression now reproduced and fixed in both languages. With $M = \mathrm{diag}(0.5, 0.5, 1)$, the correct content point is $(1720, 920)$. These independently specified numbers check origin translation, axis signs, units and viewport offsets rather than comparing a projection only against its own inverse.

5.6 Intrinsics and time-dependent geometry

For undistorted pinhole image coordinates $\tilde{q} = (u_{img}, v_{img}, 1)^T$,

$$Z_C\tilde{q} = Kp_C, \qquad K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}. \tag{14}$$

The principal point (c_x, c_y) is in image pixels and differs from the optical centre $(0,0,0)_C$. Camera intrinsics do not replace a measured gaze origin or screen registration. A pupil image point back-projects to a camera viewing ray, not automatically the visual axis of the eye. A calibrated eye model supplies that relationship. Image resizing, cropping and distortion handling must follow the calibration convention; for an affine image-coordinate change H, the corresponding intrinsic mapping is $K' = HK$. A mirrored preview must be resolved explicitly, because a single-axis reflection is not a proper 3D rotation. Content zoom modifies M, not camera intrinsics.

For a moving phone, both ray conversion and screen pose must refer to the relevant capture time. Empirical work on mobile interactions identifies head distance, head pose and device orientation as factors associated with 2D gaze-estimation error [4]. This motivates explicit geometry provenance, while the acceptable age of a pose still requires separate assessment. A matching frame name alone is insufficient. The current session uses the selected presentation's recorded pose and checks its age; the pose-age threshold does not bound spatial error from unobserved motion. The host must measure or otherwise justify its temporal geometry approximation. Orientation changes update layout epochs and physical axes/dimensions. The corrected simulation swaps one panel's physical dimensions, rather than inferring a new device size from the rotated pixel width. The desktop interaction fixture crops its viewport while keeping the active panel, top-left origin and metric pixel pitch fixed. Physical dimensions can be supplied explicitly; defaults remain synthetic assumptions.

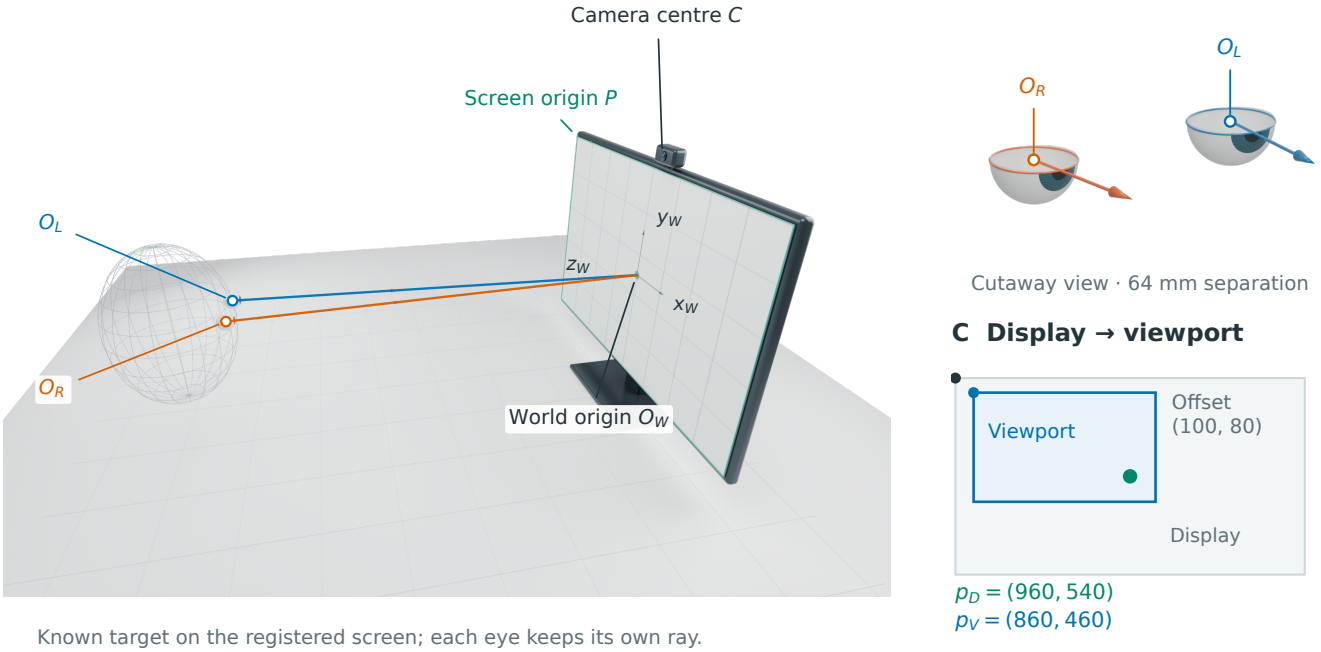


Figure 3: Independent eyes retain separate origins. (A) A procedurally rendered metric scene separates the camera centre C, active-panel origin P, world origin O_W and two eye origins O_L, O_R. Both rays meet the known centre target in this fixture without being fused. (B) A magnified cutaway makes the distinct eye centres visible; their separation is 64 mm. (C) The same screen hit maps from $(960, 540)$ in the full display to $(860, 460)$ in the offset viewport. Eye anatomy and viewing perspective are illustrative; labelled origins and ray endpoints follow the shared numeric fixture.

5.7 Content rotation and free movement

The content layer may be an image, a fixed-layout text page or a group with a stable intrinsic coordinate system. Its top-left origin remains fixed in content coordinates while the layer moves within a clipped viewport. In the viewport's right/down axes, positive θ is clockwise and the similarity transform is

$$M = \begin{bmatrix} s\cos\theta & -s\sin\theta & t_x \\ s\sin\theta & s\cos\theta & t_y \\ 0 & 0 & 1 \end{bmatrix}, \qquad p_v = sR(\theta)p_i + t. \tag{15}$$

Here $s > 0$, and t is the viewport position of the intrinsic origin. The renderer uses a top-left transform origin; an additional CSS-centred origin would change this mapping. Content rotation leaves the calibrated metric screen plane unchanged. Both PoG and screen-ray intersections use the inverse of the matrix selected at capture time.

For a viewport anchor a moving to b, first recover $q = s^{-1}R(-\theta)(a - t)$. A scale factor f and angular change δ then give

$$s' = \mathrm{clamp}(sf, s_{\min}, s_{\max}), \quad \theta' = \theta + \delta, \quad t' = b - s'R(\theta')q. \tag{16}$$

Thus q stays under the new anchor before boundary constraints. A two-contact gesture derives a, b from the initial/current midpoints, f from their distance ratio and δ from the unwrapped angular change. Scale, rotation and movement are applied together before bounding once. Near-coincident contacts suspend scale and angle changes and establish a new baseline on separation; lifting one contact rebases to one-finger movement. Desktop users can drag in a rotation mode or use bracket keys for 15° steps. The API accepts radians and movement in viewport logical pixels.

Two boundary policies serve different tasks. Free movement permits repositioning at fit scale. Bounded movement clamps the axis-aligned envelope of all four rotated corners and centres an axis when that envelope is smaller than the viewport; it does not guarantee coverage of every viewport corner. Clipping can expose blank space, so outside-content flags remain part of gaze interpretation. Reset restores zero rotation, centred content and the original unrotated fit. Independently animated children, text reflow and occlusion need additional host state; the provided layer does not infer those changes.

5.8 Observation identity and optional pairing

Observation identity includes source, clock ID and epoch, eye, representation and sample ID. The same physical capture can thus carry left/right and PoG/ray records without collision. Optional capture and calibration identifiers retain further provenance. A combined estimate stays combined; no post hoc label turns it into an independent monocular observation. Duplicate protection covers every pending observation as well as a bounded history of recent

identifiers. An identifier older than both windows can be accepted again; callers must therefore issue stable, unique sample identifiers.

For analyses that need matched eyes, an offline helper performs chronological, one-to-one pairing under a caller-specified maximum skew, defaulting to 5 ms. It considers only usable observations with matching source, representation, clock identity/epoch, surface/epoch and calibration. Unmatched observations are returned explicitly. Different source clocks can be synchronized by the host, but the helper conservatively leaves them separate. It does not implement an optimal assignment, estimate vergence or fuse measurements. Study-specific fusion requires its own uncertainty and validity policy.

6 Adapters, diagnostics and developer contract

6.1 SDK-independent normalization

Both language implementations expose generic model, Tobii Pro dictionary, EyeLink bridge and Pupil Core adapters. A generic model adapter supports nested field paths, explicit timestamp units, display-pixel normalization and metric origin conversion. Model inference and tensor transfer remain outside the core. A new adapter can be a pure function producing the shared JSON schema and structured warnings.

Tobii normalization preserves per-eye validity and normalized display points; optional rays are defined from the vendor origin to the reported display point, with millimetres converted to metres [8]. Confidence remains null when only validity is provided. The EyeLink bridge consumes serialized per-eye pixel records and millisecond device timestamps; a Python helper extracts this bridge record from a newly received PyLink sample [10]. Neither path bundles or licenses the hardware SDK on the user's behalf.

Pupil normalization preserves individual base-sample timestamps and deduplicates reused eye observations within a bounded history. Scene-camera coordinates do not become screen PoG without an explicit calibrated homography or registered surface record. Right/left IDs, bottom-origin coordinates and millimetre eye centres follow the vendor contract [9]. Mapping timestamp skew is retained and checked. These behaviors are tested with synthetic protocol fixtures; the release does not claim physical tracker validation.

6.2 Diagnostics as integration feedback

A source descriptor declares adapter, eye set, tracking mode, expected cadence and timestamp provenance. Expected cadence does not resample observations. Diagnostics estimate rates per source, eye, representation and clock epoch using bounded windows. They expose gaps, out-of-order delivery, rate deviations, missing history, uncertain clocks and unavailable geometry. Observers receive the first occurrence for each diagnostic context while the report retains counts and most recent occurrence times. An observer exception is recorded without interrupting acquisition.

Prediction reliability is distinct from clock and presentation uncertainty. Factor-Informed Uncertainty Distillation (FIUD) learns an image-quality-related error-ranking signal for selective gaze prediction [5]. Such a score is neither automatically a calibrated probability in $[0, 1]$ nor a temporal bound in milliseconds. Hosts should preserve source-specific error scores and their semantics in metadata, and retain `confidence=null` when no suitable scalar confidence is supplied.

A per-observation `usable` flag and reason list remain the primary quality contract. Informational conditions, such as unavailable scalar confidence, can coexist with usable geometry. Receipt-only timestamps, blinks, missing history or ambiguous presentation make content attention unusable. Raw measurements remain available for a different downstream analysis that states its own policy.

6.3 Human interaction and host responsibilities

The controller anchors wheel zoom at the pointer, distinguishes drag from selection-box zoom, supports keyboard reset and cancels an active gesture when another view command takes over. Touch interaction tracks contacts, pinch scale and midpoint translation; releasing one contact transitions to pan without a jump. Double tap, cancellation, resize, orientation change and background/resume are explicit cases. Browser handling follows pointer-event ownership and cancellation semantics [14]; it avoids capturing keys from embedded editable controls.

A convenience browser entry point, `createGazeSurface`, attaches one content layer to a viewport and handles input, CSS transforms, resize, lifecycle records and teardown. A runnable example groups an illustration with ordinary HTML text. Its controller exposes `rotate`, `move` and `reset`; the same operations are available in Python. The helper records the applied matrix in each render callback and preserves capture context for later tracker delivery. The host still supplies capture timestamps, source clocks and calibrated 3D geometry. This reduces integration steps, but developer usability has not yet been measured in a user study.

The lower-level browser adapter records application input, while the host records what was rendered. Native Python applications can feed the same controller logic from GUI callbacks. Both cores are synchronous; applications serialize rendering and SDK callbacks through a single session owner. No transport or thread scheduler is hidden in the measurement path. Host-specific kinetic scrolling or animation can be recorded as presentation snapshots without adding its physics to the core.

7 Verification and synthetic evaluation

7.1 Audit strategy and regression coverage

The audit follows failure paths across geometry, event ordering, browser input, recording, replay and package boundaries. Earlier defects included cross-surface frame selection, keyboard zoom overwritten by a stale drag baseline, unstable small-scale inversion, malformed-record mutation, delayed future-time quarantine and undetected replay gaps. Focused regression tests reproduce those triggers. The extension adds explicit eye identity, shared adapter fixtures, receipt-time exclusion, nullable confidence and clock-boundary checks in both languages.

The geometry audit adds independent numeric fixtures for camera/eye origins, full-display versus cropped-viewport mapping, per-eye Tobii UCS offsets and non-symmetric rotations. Both languages additionally test 100 arbitrary rigid frame changes, inverse transformations, reflected/scaled matrices, direction magnitudes spanning 10^{-200}–10^{200}, dynamic-pose provenance and physical panel dimensions across orientation changes. Five native geometry fixture

outputs are compared directly between implementations. These fixtures are separate from the 78-case acquisition-rate matrix.

Strict TypeScript compilation generates public declarations from the executable source. Consumer tests check exports and validity-dependent measurement types. Python tests exercise the independent implementation and validate normalized fixtures against JSON Schema. Release checks inspect imports, package contents, license consistency and shared fixtures. These tests establish specified behaviors under examined inputs; they are not a proof of general correctness.

7.2 Two complementary simulation families

Interaction scenarios. Desktop and mobile scenarios each last nine seconds at 60 Hz display and 120 Hz gaze, repeated with seeds 1, 41 and 90210. Desktop actions include wheel zoom, free movement, rotation, marquee selection, cancellation, reset and resize. Mobile actions include pinch in/out, two-finger twist, moving midpoint, pinch-to-pan transition, double tap, orientation and suspension. Attention policies include content-landmark tracking, screen-fixed attention during pan, scanning and reading-like fixation steps. Timing and motion parameters are illustrative test assumptions, not a fitted model of human eye movements. The moving-phone fixture holds pose between recorded frames and does not validate real continuous motion.

Rate and context matrix. A second family separates frequency sensitivity from context coverage. The controlled matrix contains 42 cells: six display rates (30, 60, 90, 120, 144 and 240 Hz) crossed with seven gaze rates per eye (30, 60, 120, 250, 600, 1,000 and 2,000 Hz). It fixes a 1280×720 viewport, 240 Hz input and independent binocular acquisition. A further 36 cases cross nine viewport configurations with left monocular, right monocular, independent binocular and combined modes. These context cases vary input rate, device pixel ratio, orientation and clock configuration together; they are coverage cases, not a factorial causal study.

The nine initial viewports range from 320×568 to 3840×2160, including portrait, landscape, tablet and ultrawide layouts. Context input rates range from 30 to 1,000 Hz. Independent context eyes use 600 and 120 Hz and separate clock mappings. Each case lasts 800 ms with seed 41. Inputs are coalesced into rendered frames. Ordinary delivery delay is 28–63 ms, with an additional 120 ms on selected captures. A blink interval occupies 46–50% of the trace. Uniform per-axis perturbation of ±0.35 logical pixels supplies an explicitly high-accuracy synthetic source; it is not a camera-model accuracy estimate.

PoG and ray observations originate from the same perturbed viewport target. Their agreement tests representation consistency rather than agreement between independent sensors. Reconstruction error is measured against that perturbed target, using the known renderer coefficients, and expressed back in logical viewport pixels. Sensor noise is therefore distinguished from software reconstruction residual. No human data are used in either simulation family.

7.3 Cross-language conformance and results

The test driver serializes identical events for both implementations and compares all aligned records and diagnostic reports. Keys, labels, frame identities and flags must match exactly. Numeric comparison uses absolute tolerance 10^{-8} plus relative tolerance 10^{-12}; the largest observed differences are much smaller (Table 3). The suite additionally compares ten source adapter fixtures and checks exact replay of derived alignment within the same implementation. Python-native experiment generation is exercised separately.

Table 3: Numerical consistency across 78 deterministic cases. Spatial errors are logical viewport pixels or content units, as indicated; they are not physical tracking accuracy.

Check	Result	Unit
Received observations	107,070	obs.
Frame mismatches	0	obs.
Exact replay	78/78	cases
Pending after drain	0	obs.
Maximum numerical differences		
Reconstruction residual	7.85×10^{-13}	px
Python–TS viewport	4.55×10^{-13}	px
Python–TS content	4.55×10^{-13}	units
Python–TS capture time	0	ms

Across the 78 cases, all 107,070 expected observations are received, pending buffers drain, and every capture-frame selection matches the independent synthetic reference. Maximum reconstruction residual is 7.85×10^{-13} logical pixels. Maximum Python–TypeScript viewport disagreement is 4.55×10^{-13} pixels; mapped capture times agree at the reported precision. Received counts include unusable observations and are not a usable-coverage statistic; reconstruction residuals are evaluated on mapped observations with a known synthetic target. These near-zero values measure arithmetic and contract consistency for known inputs. They cannot establish subpixel physical accuracy.

Additional content-motion cases. Eight further cases combine rotation and free movement in 390×844 and 1920×1080 viewports with the four eye modes. Mobile/desktop configurations use 90/144 Hz presentation and 125/1,000 Hz input, with 600 Hz primary and 120 Hz right-eye sampling where applicable. The deterministic rotation reaches ±0.8 rad; the content centre moves by up to 12% of viewport width and 10% of height while scale varies. These cases retain the 800 ms duration, noise and delivery model above; mobile cases also change display orientation. All 6,516 expected observations are received, every capture-frame choice matches, and replay is exact. Maximum mapped reconstruction residual is 8.49×10^{-13} logical pixels. These eight cases are reported separately from the 78-case baseline and are additional software consistency checks.

The receipt-time baseline's P95 residual ranges from 54.4 to 79.6 logical pixels across the controlled cells (Figure 4A). This range belongs to the specified fast manipulation trace, viewport and delays. It is not a general error bound or a comparison between tracker brands. Increasing gaze rate cannot repair an incorrect temporal association. Context-case baseline values are not pooled to infer frequency effects because dimensions and motions also differ.

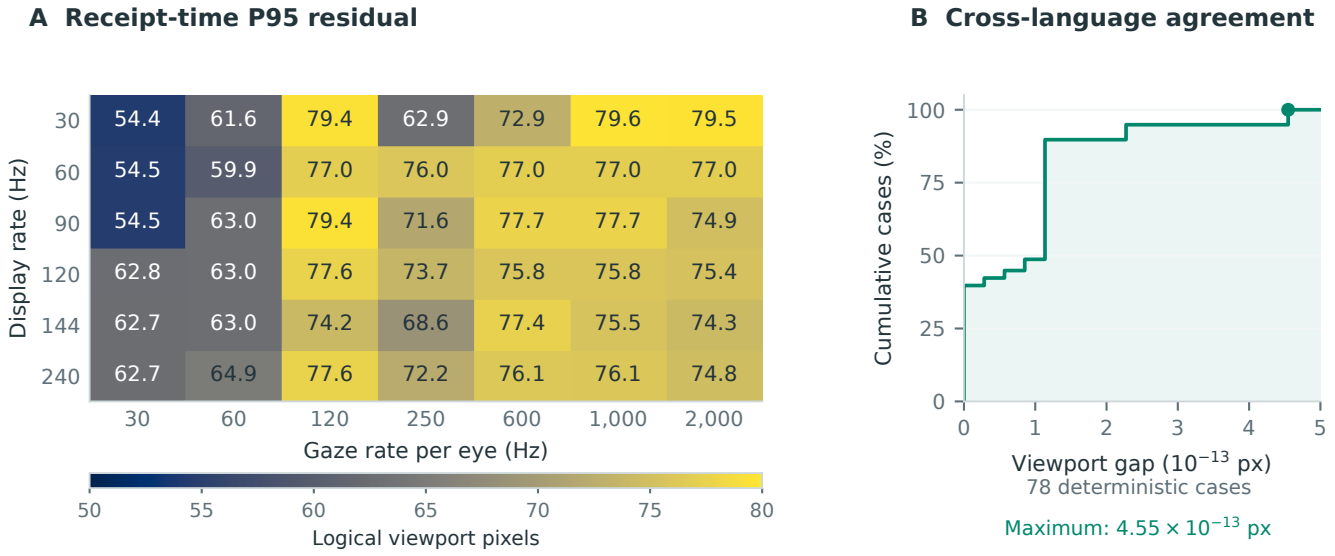


Figure 4: Controlled residuals and implementation agreement. (A) Receipt-time baseline P95 residual for the 42 rate cells at fixed 1280 × 720 viewport and 240 Hz input. (B) Empirical cumulative distribution of the maximum Python–TypeScript viewport discrepancy in each of the 78 cases, including exact-zero cases. The largest gap is 4.55×10^{-13} logical pixels. Each case is deterministic; these results establish numerical consistency, not tracker accuracy.

8 Reproducibility and distribution

The repository contains strict TypeScript source, Python source, JSON schemas, synthetic fixtures, regression tests, a local interaction lab and figure scripts. The two libraries have no runtime dependencies. The visual assets use a separate procedural figure pipeline; an editable Blender scene and projected annotation anchors reproduce the 3D geometry figure without external meshes or photographs. npm ships executable modules, generated declarations and corresponding TypeScript source. Python ships a wheel and source distribution; SDKs, inference weights and private-project data are excluded. Clean installation tests run outside the checkout, including a wheel rebuilt from the source distribution.

Schema 2.1 records the implementation version, plane extent, metric hit frame, source descriptors, clock mappings, configuration, raw event sequence, retention losses and derived output. Exact replay rejects incomplete or incompatible logs, including earlier implementation versions; recordings are never silently reinterpreted under changed projection semantics. Default in-memory limits are explicit: 7,200 frames, 100,000 raw/derived records and 4,096 pending observations. Long sessions require persistence before rollover. Python raw-log storage uses a deque to avoid shifting the retained history on each insertion. Its public snapshots and JSON exports remain ordered lists. Other bounded arrays still incur front-removal costs; streaming storage and backpressure remain measured integration work rather than claimed features.

The full matrix can be reproduced with the following source-checkout commands:

```
npm run test:conformance
node tools/paper-data.mjs
.venv/bin/python tools/paper_figures.py
```

Figure generation additionally requires the pinned plotting dependencies listed in the paper build guide. The reported local run used Node.js 22.16.0 and Python 3.12.14. Continuous integration is configured across Node 22/24/26 and Python 3.10/3.12/3.14. The aggregate JSON data preserve every configuration and result; generated working sessions remain excluded from source control.

Code is licensed GPL-3.0-only. The manuscript has separate draft publication rights. The repository is currently private and registry publishing is disabled; no public package availability or implementation arXiv identifier is claimed. Maintainer-reviewed workflows prepare the official npm and PyPI routes. Citation metadata requests both the original GazeSync paper and this implementation report; it will be updated with the real preprint identifier after submission.

9 Limitations and evaluation agenda

End-to-end assessment must distinguish estimation and calibration quality, capture-to-presentation association, and coordinate reconstruction under the registered geometry. This validation tests software consistency under supplied clock mappings and synthetic geometry. Physical exposure time, display scanout, clock offset and drift, extrinsic calibration, source accuracy and moving-device pose remain integration measurements. Render callbacks alone do not measure photon onset. Replaying known geometry cannot validate an assumed camera-to-display transform. Protocol fixtures cannot establish behavior across firmware or SDK versions, and a phone-sized browser simulation cannot certify touch behavior on every mobile browser.

The next evaluation should measure timing against an external reference, report latency distributions and uncertainty, and test display geometry on held-out targets. Static views, pan, zoom, rotation and device motion should be assessed separately before

compound manipulations. Relevant outcomes include temporal ambiguity, usable coverage, per-eye dropout, viewport/content error and angular ray error. Error on retained observations should be reported together with the proportion retained and the reasons for exclusion, so rejection cannot conceal poor coverage. Calibration and assessment targets must remain separate. Repeated sessions and participant-separated analysis are needed to characterize drift and generalization.

Human behavior also needs empirical grounding. The scripted attention policies exercise distinct reconstruction cases but do not estimate how often people follow a pointer, retain a landmark or look ahead of an action. A future study should record task intent and interaction phases alongside independent gaze, then test whether the instrumentation preserves meaningful content attention without changing normal interaction. Longer runs should measure memory, persistence overhead and worst-case latency under realistic acquisition load.

10 Conclusion

GazeSync provides an explicit bridge between heterogeneous eye-tracking records and the content state visible during desktop or mobile interaction. Its temporal contract, geometric conventions, independent eye records and quality flags make assumptions inspectable. Matching Python and TypeScript implementations, adapter fixtures and synthetic experiments establish a reproducible software foundation. Measured device timing, calibration and participant behavior remain the necessary next evidence for end-to-end claims.

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